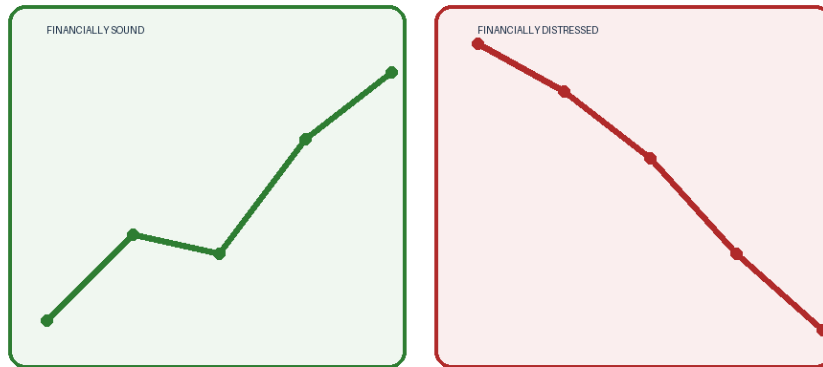


A COMPARATIVE STUDY OF

WEIGHTED AVERAGE COST OF CAPITAL & WORKING CAPITAL MANAGEMENT

A Financially Sound Company vs. a Financially Distressed Company



JSW Steel Limited vs. **Housing Development & Infrastructure Ltd**

A Research Report

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Abstract

This report undertakes a comparative analysis of two Indian listed companies representing opposite ends of financial performance: **JSW Steel Limited**, a financially sound, integrated steel producer, and Housing Development and Infrastructure Limited (**HDIL**), a real-estate company currently under Corporate Insolvency Resolution Process (CIRP) with a deeply negative net worth. The study estimates and compares the Weighted Average Cost of Capital (**WACC**) of each company and evaluates their working capital management practices using standard efficiency measures — Days Sales Outstanding (**DSO**), Days Payable Outstanding (**DPO**) and Inventory Turnover Days. The findings show that JSW Steel's WACC of 7.60% reflects a diversified, market-priced capital structure with genuine debt servicing obligations, while HDIL's WACC of 6.60% is a statistical artefact of a distressed, thinly-traded stock with near-zero beta and unserviced debt, rather than a sign of cheap capital. On working capital, JSW Steel maintains efficient, industry-typical cycles (DSO of approximately 18 days, inventory days of about 127), whereas HDIL's working capital ratios are extreme and economically meaningless (DSO exceeding 350 days) due to its near-total loss of operating revenue. The report concludes that WACC and working-capital ratios must always be interpreted alongside the qualitative financial health of a company, since formulaic outputs can be misleading for a distressed entity.

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Chapter 1 — Introduction

1.1 Background of the Study

Two financial pillars determine the long-term health of any company: the cost at which it raises capital, and the efficiency with which it manages its short-term operating cycle. The Weighted Average Cost of Capital (WACC) represents the blended return that a firm must earn on its investments to satisfy both its equity shareholders and its debt holders, and is widely used as the discount rate in valuation and capital budgeting decisions (Damodaran, 2012). Working Capital Management, on the other hand, concerns the day-to-day management of current assets and current liabilities — inventory, receivables, payables and cash — and directly affects a firm's liquidity and operational efficiency (Deloof, 2003).

This study contrasts these two dimensions across two Indian listed companies that sit at opposite ends of the financial-health spectrum: JSW Steel Limited, one of India's largest and most profitable integrated steel producers, and Housing Development and Infrastructure Limited (HDIL), a former large-cap real-estate developer that has been under Corporate Insolvency Resolution Process (CIRP) since August 2019, with auditors issuing a Disclaimer of Opinion on its most recent financial statements.

1.2 Problem Statement

Standard financial formulas for WACC and working-capital ratios are built on assumptions of a going concern with actively traded securities and normal operating activity. When applied mechanically to a financially distressed company, these formulas can produce numbers that are technically correct but economically misleading — for example, a very low WACC arising not from genuine low risk but from a collapsed, thinly-traded share price and unserviced debt. This report examines whether, and how, the WACC and working-capital metrics of a distressed company (HDIL) diverge in meaning from those of a

healthy company (JSW Steel), even when calculated using identical methodology.

1.3 Research Objectives

- To estimate the Weighted Average Cost of Capital (WACC) of JSW Steel Limited and HDIL using their respective capital structures, cost of equity and cost of debt.
- To evaluate and compare the working capital management practices of both companies using DSO, DPO, inventory turnover days and net working capital.
- To identify how financial distress distorts the interpretation of standard cost-of-capital and working-capital metrics.
- To draw practical lessons for financial analysts and investors on interpreting WACC and working-capital ratios of distressed companies.

1.4 Research Questions

- What is the estimated WACC of JSW Steel Limited and HDIL, and what factors drive the difference between them?
- How do the working capital cycles (DSO, DPO, inventory days) of a financially sound company compare with those of a financially distressed company?
- Do lower WACC figures for a distressed company genuinely represent a lower cost of capital, or a statistical distortion?

1.5 Hypotheses

- H1: There is a significant difference between the WACC of a financially sound company and a financially distressed company.
- H2: There is a significant difference in working capital efficiency (DSO, DPO and inventory turnover days) between a financially sound company and a financially distressed company.
- H3: A financially distressed company's WACC understates its true cost of capital due to depressed share price and non-servicing of debt.

1.6 Significance of the Study

This comparative study is useful to investors, credit analysts and finance students in understanding the practical limitations of applying textbook WACC and working-capital formulas to distressed companies. It also illustrates, through a real, publicly disclosed pair of companies, how insolvency and going-concern doubt manifest in financial ratios.

1.7 Scope and Limitations

- The study is limited to two companies — JSW Steel Limited and Housing Development and Infrastructure Limited — and is not intended to generalise across the steel or real-estate sectors as a whole.

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- Financial data used is drawn from the companies' published financial statements (FY 2023-24 and FY 2024-25) and publicly available market data (NSE, BSE, moneycontrol.com, Bloomberg).
 - HDIL's data reflects a company under CIRP; its ratios are affected by suspended/limited operations and should not be read as representative of a normally functioning real-estate company.
 - The study does not perform independent audit verification of the underlying financial statements.

1.8 Definition of Key Terms

- WACC (Weighted Average Cost of Capital): The average rate of return a company must pay its security holders (equity and debt), weighted by their proportion in the capital structure.
- Cost of Equity (K_e): The return required by equity shareholders, estimated here using CAPM (HDIL) and the Dividend Growth Model (JSW Steel).
- Cost of Debt (K_d): The effective interest rate a company pays on its borrowings, net of the tax shield.
- Working Capital: The difference between a company's current assets and current liabilities, indicating its short-term liquidity.
- DSO / DPO: Days Sales Outstanding and Days Payable Outstanding — the average number of days a company takes to collect receivables and pay its suppliers, respectively.
- CIRP: Corporate Insolvency Resolution Process, a statutory process under India's Insolvency and Bankruptcy Code, 2016.

Chapter 2 — Literature Review

The concept of the Weighted Average Cost of Capital is rooted in the capital structure theory of Modigliani and Miller (1958), which established the relationship between a firm's cost of capital, its capital structure and its overall value. Damodaran (2012) extended this into practical estimation techniques, including the use of comparable-company and CAPM-based approaches for firms with limited or unreliable trading history — an approach directly relevant to a distressed, thinly-traded company such as HDIL in this study.

The Capital Asset Pricing Model (CAPM), developed by Sharpe (1964) and Lintner (1965), remains the most widely used framework for estimating the cost of equity, linking expected return to a company's systematic risk (beta) relative to the market. However, several studies note that CAPM-derived betas for distressed or illiquid stocks can be statistically unreliable, often understating true risk because price movements become disconnected from fundamentals (Fama and French, 1992).

On working capital, Deloof (2003) demonstrated a significant relationship between working capital management efficiency and corporate profitability among Belgian firms, finding that reducing receivables and inventory days is generally associated with higher profitability. Richards and Laughlin (1980) introduced the Cash Conversion Cycle as a comprehensive measure of working capital efficiency, integrating inventory, receivables and payables periods into a single metric. Padachi (2006) similarly found that the composition and efficiency of working capital significantly affects small manufacturing firms' performance.

Studies on corporate distress, including Altman's (1968) Z-Score model, highlight that financially distressed firms typically display a combination of negative or eroding net worth, poor liquidity and declining operating cash flow — all of which are directly observable in HDIL's financial statements, including its disclaimer of audit opinion and status under the Insolvency and Bankruptcy Code. Collectively, the literature suggests that while WACC and working-capital ratios are theoretically robust tools, their outputs must be interpreted cautiously — and in the context of firm-specific solvency and liquidity conditions — rather than compared at face value across companies in fundamentally different financial states.

Chapter 3 — Research Methodology

3.1 Research Design

This study adopts a descriptive, comparative research design based on secondary financial data. Two contrasting companies were purposively selected to represent a financially sound company (JSW Steel Limited) and a financially distressed company (Housing Development and Infrastructure Limited).

3.2 Population and Sample

The population comprises publicly listed Indian companies with available audited financial statements. The sample consists of two companies, selected purposively: JSW Steel Limited (NSE: JSWSTEEL) representing a financially healthy, going-concern company, and HDIL (BSE: HDIL) representing a financially distressed company under CIRP.

3.3 Data Collection

Secondary data was collected from the audited standalone and consolidated financial statements (Balance Sheet, Statement of Profit & Loss, and Cash Flow Statement) of both companies for FY 2023-24 and FY 2024-25, sourced from the companies' published disclosures. Market data (share price history, dividend history) was sourced from NSE India, BSE India, Moneycontrol and Bloomberg.

3.4 Research Instrument

A structured financial-analysis worksheet was used to compute the WACC and working-capital ratios for both companies on a consistent basis, using the formulas described below.

3.5 Data Analysis Methods

Weighted Average Cost of Capital (WACC)

- $WACC = (K_e \times E/V) + (K_d(1-T) \times D/V)$, where E and D are the market/book values of equity and debt, $V = E + D$, T is the corporate tax rate.
- Cost of Equity — JSW Steel: estimated using the Dividend Growth (Gordon) Model: $K_e = (D_1 / P_0) + g$, where D1 is the expected next dividend, P0 is the current market price and g is the historical dividend growth rate.
- Cost of Equity — HDIL: estimated using CAPM: $K_e = R_f + \beta(R_m - R_f)$, where R_f is the risk-free rate, β is the regression beta of the stock against the Nifty 50, and R_m is the market return.
- Cost of Debt: $K_d(\text{after tax}) = \text{Pre-tax cost of debt} \times (1 - \text{Tax Rate})$, based on effective interest cost on total borrowings.

Working Capital Management

- Days Sales Outstanding (DSO) = (Trade Receivables / Revenue from Operations) × 360
- Days Payable Outstanding (DPO) = (Trade Payables / COGS) × 360
- Inventory Turnover (Days) = (Inventories / COGS) × 360
- Net Working Capital = Current Assets – Current Liabilities (cash, trade receivables and trade payables considered)

3.6 Ethical Considerations

All data used in this study is drawn from publicly available, company-published financial statements and recognized financial data platforms. No confidential or non-public information has been used. HDIL's distressed status and the disclaimer of audit opinion issued by its auditors are disclosed transparently in this report, in line with its public filings.

Chapter 4 — Data Analysis and Results

4.1 Company Snapshots

	JSW Steel Limited	HDIL
Sector	Steel manufacturing	Real estate & housing infrastructure
Status	Going concern, profitable	Under CIRP since Aug 2019; Disclaimer of Opinion
Revenue FY25 (₹ Cr)	1,68,824 (consol.)	33.96
PAT FY25 (₹ Cr)	5,837 (standalone)	(0.01) — near-nil loss
Net Worth	Positive; ₹79,496 Cr equity base	Deeply negative (Other Equity deficit of ~₹2,769 Cr)
Share Price Trend	Market leader, actively traded	Collapsed from ₹91.45 (2016) to ₹3.20 (2025), ~96% decline

Figures for JSW Steel are ₹ in Crores original HDIL figures are ₹ in Lakhs and have been converted to Crores here for comparability (1 Cr = 100 Lakhs).

4.2 Weighted Average Cost of Capital (WACC)

4.2.1 JSW Steel Limited

Component	Value	Basis
Total Equity	₹79,496 Cr	Book value
Total Debt	₹98,752 Cr	Borrowings + lease liabilities
Total Capital	₹1,78,248 Cr	Equity + Debt
Equity Weight	44.60%	
Debt Weight	55.40%	
Cost of Equity (Ke)	9.63%	Dividend Growth Model: D1 ₹3.06, P0 ₹1,233.70, g 9.38%
Pre-Tax Cost of Debt	8.52%	Effective interest rate
Tax Rate	30.00%	
After-Tax Cost of Debt (Kd)	5.96%	
WACC	7.60%	

4.2.2 Housing Development & Infrastructure Ltd (HDIL)

Component	Value	Basis
Total Equity (paid-up capital)	₹474.00 Cr	Equity share capital only — net worth is negative
Total Debt (formal)	₹0.00 Cr	No interest currently being serviced under CIRP
Equity Weight	100.00%	
Debt Weight	0.00%	
Beta (β)	0.038	Regression vs Nifty 50, annual closing prices, 2016–2025
Risk-Free Rate (R_f)	6.38%	10-year G-Sec (Bloomberg)
Market Return (R_m)	12.23%	Nifty 50, ~9-year CAGR
Cost of Equity (CAPM)	6.60%	$K_e = R_f + \beta(R_m - R_f)$
Cost of Debt (after tax)	0.00%	Interest not serviced under CIRP
WACC	6.60%	

HDIL's Current Borrowings of ₹2,483.64 Cr appear on its balance sheet but are excluded from the formal cost-of-debt calculation, as the company is not servicing interest on this debt while under CIRP. HDIL's near-zero beta reflects a largely stagnant, thinly-traded share price rather than genuinely low risk.

4.3 Working Capital Management

4.3.1 JSW Steel Limited (Consolidated, ₹ Cr)

Particulars	FY 2024-25	FY 2023-24
Revenue from Operations	1,68,824	1,75,006
COGS	99,142	1,01,678
Cash & Cash Equivalents	11,655	8,030
Inventories	34,956	37,815
Trade Receivables	8,415	7,548
Trade Payables	12,018	15,711

Particulars	FY 2024-25	FY 2023-24
Dys Sales Outstanding (DSO)	17.94 days	15.53 days
Days Payable Outstanding (DPO)	25.63 days	32.32 days
Inventory Turnover (Days)	126.93 days	133.89 days
Net Working Capital	₹43,008 Cr	₹37,682 Cr

4.3.2 HDIL (₹ Cr, converted from Lakhs)

Particulars	FY 2024-25	FY 2023-24
Revenue from Operations	33.96	2.05
COGS	272.73	(76.20)
Cash & Cash Equivalents	16.51	17.02
Inventories	2,814.69	2,837.44
Trade Receivables	406.34	398.71
Trade Payables	418.40	417.49
Days Sales Outstanding (DSO)	359.0 days	5,846.5 days
Days Payable Outstanding (DPO)	369.7 days	6,121.9 days
Inventory Turnover (Days)	309.6 days	(1,117.1) days
Net Working Capital	₹2,819.13 Cr	₹2,835.68 Cr

HDIL's DSO, DPO and inventory-day figures are extreme (300+ to 6,000+ days) because the ratios are calculated against a near-zero revenue and COGS base — the company has effectively ceased normal operations. These figures should be read as evidence of distress, not as meaningful operating-cycle benchmarks.

4.4 Summary Comparison

Metric	JSW Steel (Good)	HDIL (Bad)
WACC	7.60%	6.60%
Cost of Equity	9.63%	6.60%
Cost of Debt (after tax)	5.96%	0.00%
DSO	~18 days	~359 days
Inventory Days	~127 days	~310 days
Net Worth	Positive, growing	Deeply negative
5-yr+ Share Price Trend	Positive / growth	Negative (~-96% since 2016)

Chapter 5 — Discussion

5.1 Interpreting the WACC Comparison

At first glance, HDIL's WACC of 6.60% appears lower — and therefore more attractive — than JSW Steel's 7.60%. This is a textbook illustration of why WACC cannot be compared at face value across companies in different financial states. JSW Steel's WACC reflects a real, market-tested capital structure: a meaningful proportion of debt (55.4%) on which it pays a genuine market interest rate (8.52% pre-tax), and an equity cost derived from an actively traded, dividend-paying stock. HDIL's lower figure, in contrast, arises mechanically from (a) a beta of just 0.038 — the result of a stock that has been largely frozen or thinly traded rather than genuinely low-risk, and (b) a 0% cost of debt that reflects unserviced borrowings under insolvency proceedings, not an absence of financial obligation. In substance, HDIL's true cost of capital — and its risk to any investor — is far higher than the 6.60% figure suggests.

5.2 Interpreting the Working Capital Comparison

JSW Steel's working capital cycle is consistent with a well-run, large-scale manufacturer: receivables are collected in around 18 days, and although inventory days (~127) are relatively high — typical for a capital- and inventory-intensive steel business — the company comfortably finances this with supplier credit and internal cash generation, evidenced by growing net working capital of ₹43,008 Cr. HDIL's working capital ratios, on the other hand, are not economically interpretable in the conventional sense. A DSO of 359 days (and 5,846 days in the prior year) does not indicate poor collections practice so much as a company whose revenue base has collapsed to near-zero, making any ratio with revenue in the denominator extremely volatile and, in the FY24 case, close to meaningless.

5.3 Testing the Hypotheses

- H1 (WACC difference): Supported on the surface (7.60% vs 6.60%), but the direction is counter-intuitive — the distressed company shows a lower headline WACC, driven by distortion rather than lower risk. This nuance is important and is discussed further below.
- H2 (Working capital efficiency difference): Strongly supported — JSW Steel's working capital cycle is between 15–20x shorter than HDIL's across DSO, DPO and inventory days.
- H3 (Understated cost of capital for distressed company): Supported — HDIL's WACC understates its true cost of capital, given its near-zero beta (a product of illiquid trading) and zero cost of debt (a product of non-payment, not favourable financing terms).

5.4 Broader Implications

This comparison illustrates a practical caution for analysts and investors: WACC and working-capital ratios are diagnostic tools that assume operational and financial normalcy. When a company is under insolvency proceedings, as HDIL is, these metrics should be supplemented with qualitative indicators — audit opinions, going-concern notes, net worth trends and share price history — before being used for valuation or investment decisions.

Chapter 6 — Conclusion and Recommendations

6.1 Conclusion

This study compared the Weighted Average Cost of Capital and working capital management of JSW Steel Limited, a financially sound market leader, and HDIL, a company under insolvency resolution. JSW Steel's WACC of 7.60% and efficient working capital cycle (DSO ~18 days, inventory days ~127) reflect a healthy, well-capitalised business. HDIL's WACC of 6.60% and extreme working-capital ratios (DSO of 359+ days) reflect not a low-risk, efficient business, but a company whose formal financial metrics have been distorted by insolvency, negligible revenue and a collapsed share price. The comparison confirms that both WACC and working-capital ratios must be read in context — a lower WACC is not automatically 'better' if it results from financial distress rather than genuine low risk.

6.2 Recommendations

- Investors and analysts should never rely on a company's headline WACC in isolation; it should always be cross-checked against qualitative indicators such as audit opinion, net worth trend and going-concern status.
- For companies under CIRP or facing distress, analysts should use adjusted or scenario-based cost-of-capital estimates rather than mechanically computed CAPM/dividend-model outputs, since betas and dividend growth rates become unreliable in illiquid or non-paying scenarios.
- JSW Steel should continue to monitor its inventory days, which — while typical for the steel sector — represent the largest component of its working capital and the greatest opportunity for efficiency gains.
- HDIL's case reinforces the importance, for lenders and regulators, of early warning indicators (falling share price, rising payable/receivable days, negative net worth) that can flag distress well before formal insolvency proceedings begin.

6.3 Scope for Future Research

Future studies could extend this comparison to a larger sample of financially sound and distressed companies within the same sector, to test whether the patterns observed here — a distorted, artificially low WACC and extreme working-capital ratios under distress — hold more generally. A sector-matched comparison (e.g., two real-estate companies, one healthy and one distressed) could further isolate the effect of financial distress from industry-specific working-capital norms.

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